DEEP LEARNING

Report 5: Backpropagation

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Implementation

Feedforward
The part is done in labwork 4. This part will process inputs layer by layer, calculating the weighted sum z and activation a for each neuron.

$$z = \sum_{i=1}^{n} x_i w_i + b$$

All neurons use the sigmoid activation:

$$a = \sigma(z) = \frac{1}{1 + e^{-z}}$$

Backpropagation
Training relies on minimizing the Cross-Entropy loss function :

$$J = -\frac{1}{N} \sum_{i=1}^{N} [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)]$$

The backpropagation will compute the error in the output layer and propagate it backward to update the weights and biases of each neuron. The error signal δ for each neuron is calculated based on the derivative of the loss function with respect to the neuron's output, and then used to update the parameters.

- **Output layer:** $\delta = \hat{y} - y$
- **Hidden layers:** $\delta_j = (\sum_k \delta_k w_{kj}) \cdot \hat{y}_j(1 - \hat{y}_j)$

where w_{kj} is the weight from hidden neuron j to neuron k in the next layer, and δ_k is the error signal for neuron k in the next layer.
The weights and biases are updated using the calculated error signals:

$$w = w - lr \times \delta \times x, \qquad b = b - lr \times \delta$$

Training Configuration
The optimization of training loop use a reusable function to compute the gradient descent and the loss. The function will take two callbacks as arguments: one for computing the loss and another for computing the gradients.
The learning rate is set to 0.5, and the training will stop when the loss improvement is less than $1e-6$ or after a maximum of 10000 epochs.

XOR Experiment

The model successfully converged at epoch 4000/10000 with the detailed results as follows:

Table 1: XOR Truth Table vs Model Predictions

X_1	X_2	y	$\hat{y}$
0	0	0	0.002586
0	1	1	0.998124
1	0	1	0.998118
1	1	0	0.002663